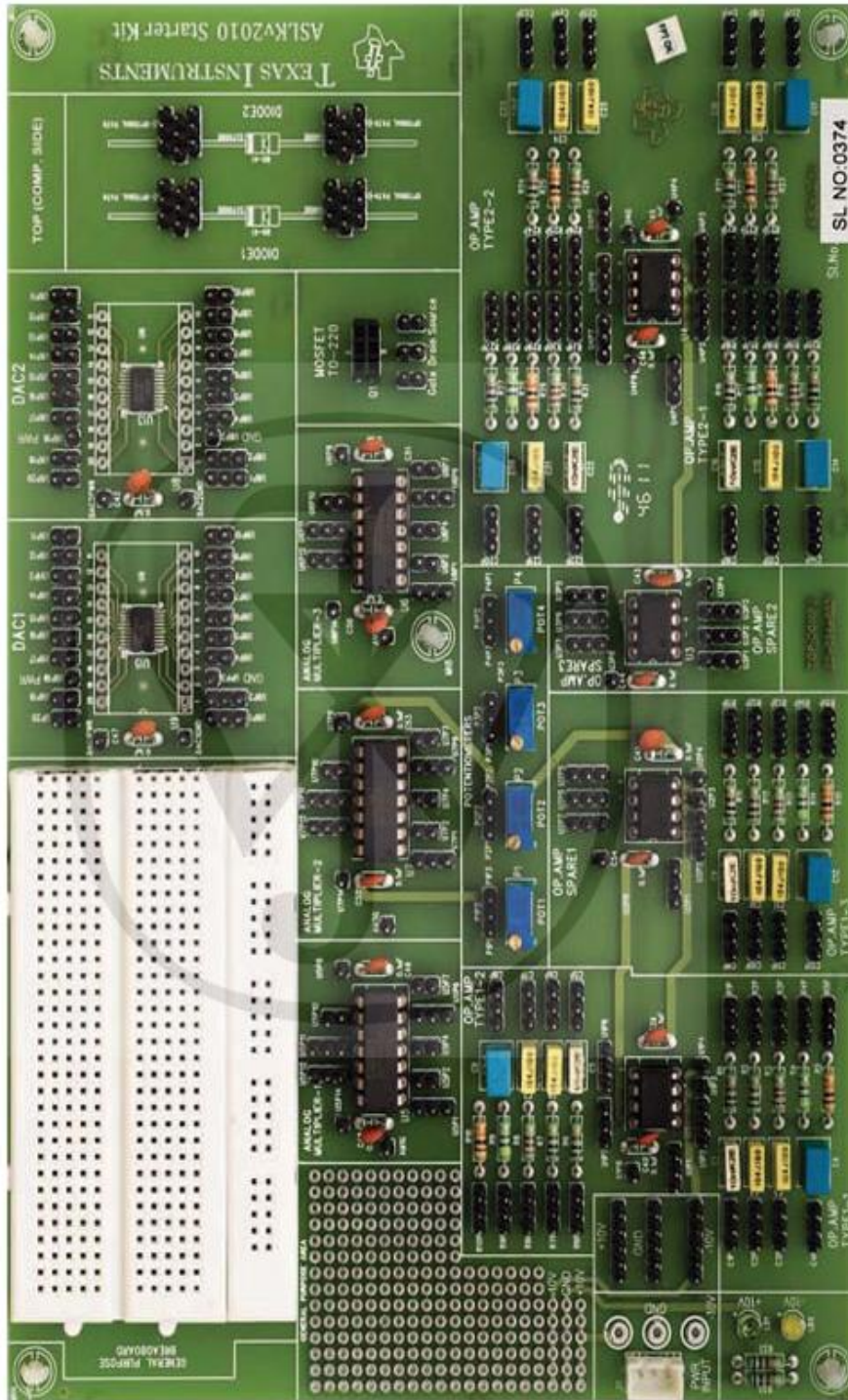


Explore the fascinating world of **Analog System Design** with **ASLKV2010 STARTER KIT**



Pin Diagram for ASLKv2010 Starter Kit

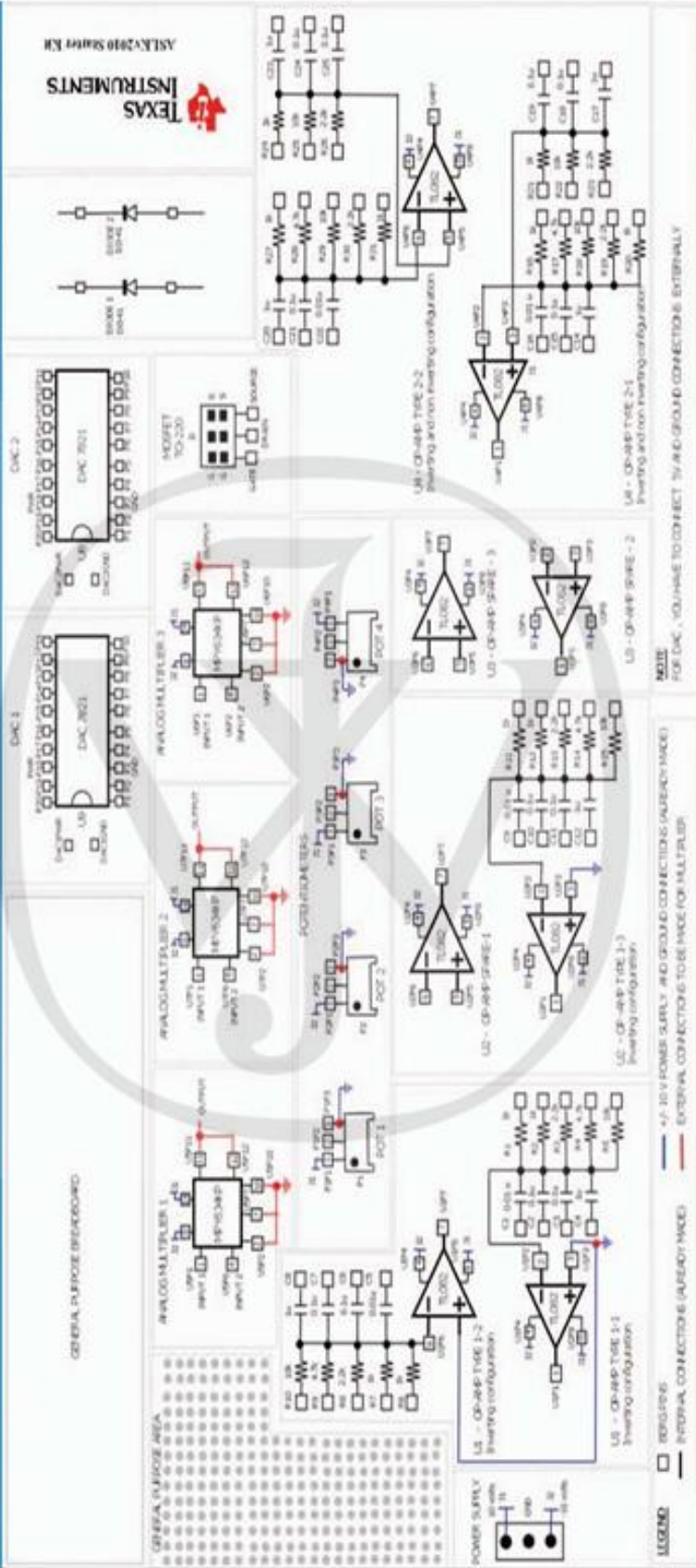


Figure 0-4 Pin diagram for ASLKv2010 Starter kit

0.4 System Lab Kit ASLKv2010 Starter: An Overview

0.4.1 | Hardware

ASLKv2010 Starter kit (see Figure 0-3; Pin diagram is shown in Figure 0-4) has been developed at Texas Instruments, India. This kit is designed for undergraduate

Introduction

engineering students to perform analog lab experiments. The main idea behind ASLKv2010 Starter kit is to provide a cost-efficient platform or test bed for students to realize almost any analog system using general-purpose ICs such as Op-Amps and analog multipliers.

ASLKv2010 Starter kit comes with four general-purpose operational amplifiers (TL082) and three wide-bandwidth precision analog multipliers (MPY634) from Texas Instruments. There is also a provision to include a 12-bit parallel-input multiplying digital-to-analog converter DAC7821. A portion of ASLKv2010 Starter kit is left for general-purpose prototyping and can be used for carrying out mini-projects.

The kit has a provision to connect ± 10 V DC power supplies. The kit comes with the necessary short and long connectors, as well as connectors for power supplies. Figure D-2 (Appendix D) shows the way power supply connections are made on ASLKv2010 Starter kit. The ± 10 V supplies are connected internally to all the ICs that require ± 10 V supplies, namely, the operational amplifiers and the multipliers. Since the DAC requires 5 V supply, the student has three options – to use the output from one of the potmeters to provide 5 V supply, to generate a 5 V supply (see the DC-DC converter experiment in this manual) or to use an external 5 V supply.

This comprehensive user manual included with the kit gives complete insight of how to use ASLKv2010 Starter kit. The manual covers exercises of analog system design along with brief theory and simulation results obtained using simulation software.

Refer to Appendix A for the details of the integrated circuits that are included in ASLKv2010 Starter kit. Refer to Appendix D for additional details of ASLKv2010 Starter kit.

0.5 Familiarizing with ASLKv2010 Starter Kit

The **Analog System Lab ASLKv2010 Starter** kit is divided into many sections. Refer to the picture in Figure 0-5 when you read the following description. Please also refer to the floorplan of the chip shown in Figure D-1 (Appendix D).

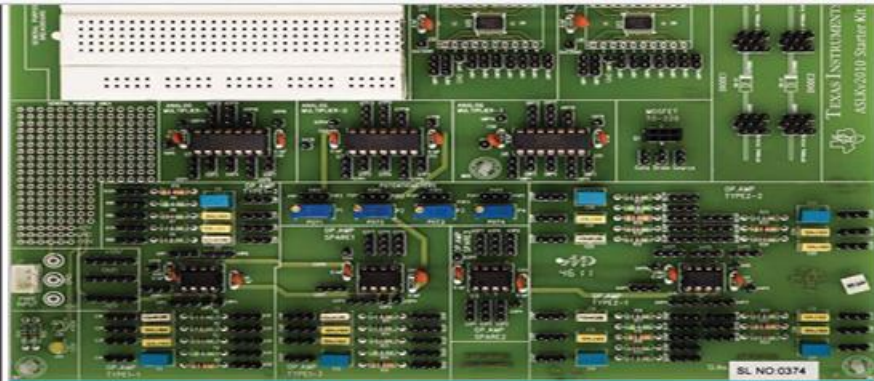
Figure D-1 shows the overall floorplan of the ASLKv2010 Starter kit. We have shown the power connections in ASLKv2010 Starter in Figure D-2. Note that the ± 10 V power and ground connections have to be connected to the power inlets at the side of the kit; the power and ground are internally connected to the Op-Amps and analog multipliers. If you wish to carry out an experiment using the DAC integrated

circuits on the board, you must use an external 5 V supply and ground connection, as shown in Figure D-2. There are three potmeters included in the kit, each of which is connected across 10 V and ground. The output of the potmeter can be used to derive a voltage in the range of 0 to 10 V; this can be useful in generating a reference voltage or even in generating a 5 V power supply for the DAC.

- There are four TL082 Op-Amp ICs labeled 1, 2, 3 and 4 on ASLKv2010 Starter kit. Each of these ICs has two amplifiers, which are labeled A and B. Thus, 1A and 1B are the two Op-Amps in the Op-Amp IC 1, etc. The eight Op-Amps are categorized as shown in the following table:

Op-Amp IC	Op-Amp	Label on Kit	Type	Purpose
1	1A	TYPE 1-1	TYPE-1	Inverting Configuration only
	1B	TYPE 1-2	TYPE-1	Inverting Configuration only
2	2A	TYPE 1-3	TYPE-1	Inverting Configuration only
	2B	TYPE SPARE-1	SPARE	Spare
3	3A	TYPE SPARE-2	SPARE	Spare
	3B	TYPE SPARE-3	SPARE	Spare
4	4A	TYPE 2-1	TYPE-2	Inverting or Non-inverting
	4B	TYPE 2-2	TYPE-2	Inverting or Non-inverting

Refer to the floorplan of the kit and identify the Op-Amp ICs (Figure D-1). It will also be helpful to refer to the power connections shown in Figure D-2. Please see connection diagrams shown in Figures D-3, D-4, D-5 and D-6. The Op-Amps are marked TYPE 1, TYPE 2 or SPARE on the board. The Op-Amps marked TYPE 1 can be connected in the inverting configuration (only). With the help of connectors, either resistors or capacitors can be used in the feedback loop of the amplifier. There are three TYPE 1 amplifiers. There are three spare Op-Amps and two TYPE-2 amplifiers. TYPE-2 amplifiers can be connected in inverting or non-inverting configurations. All the Op-Amps ICs operate on ± 10 V; the power supply and ground connections are internally provided and the user need not worry about these.



ICs used in ASLKv2010 Starter Kit

Texas Instruments Analog ICs used in ASLKv2010 Starter kit

A.1 TL082: JFET-Input Operational Amplifier

A.1.1 | Features

- ❶ Low power consumption.
- ❷ Wide common-mode and differential voltage ranges.
- ❸ Input bias and offset currents.
- ❹ Output short-circuit protection.
- ❺ Low total harmonic distortion: . . . 0.003% Typ.
- ❻ High input impedance: . . . JFET-input stage.
- ❼ Latch-up-free operation.

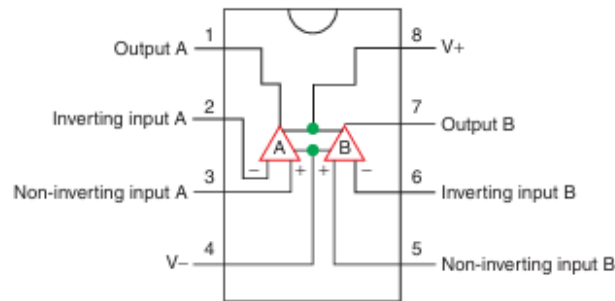


Figure A-1 TL082 – JFET-input operational amplifier

- ⑧ High slew rate: ... 13 V/ μ s Typ.
- ⑨ Common-mode input voltage range includes V_{CC+} .

A.1.2 | Applications

- ① Instrumentation Amplifiers
- ② Filters

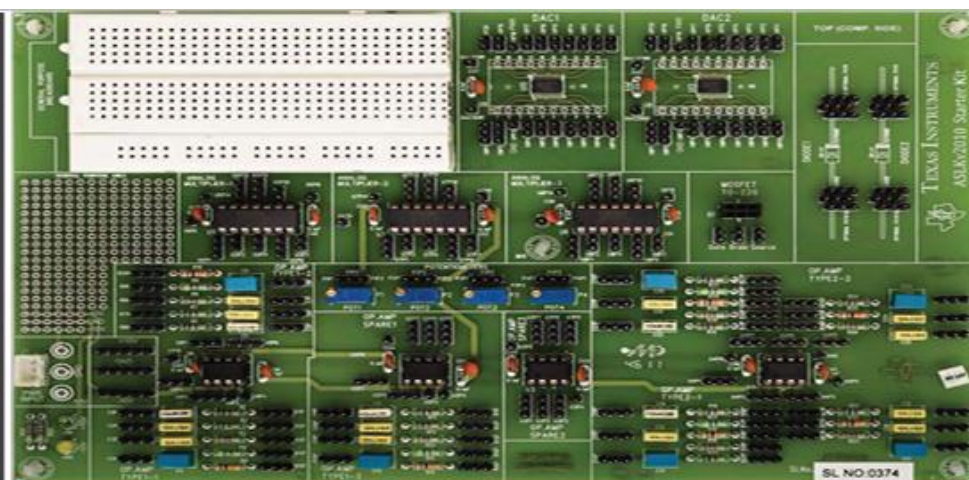
A.1.3 | Description

The TL08x JFET-input operational amplifier family is designed to offer a wider selection than any previously developed operational amplifier family. Each of these JFET-input operational amplifiers incorporates well-matched, high-voltage JFET and bipolar transistors in a monolithic integrated circuit. The devices feature high slew rates, low input bias and offset currents, and low offset voltage temperature coefficient. Offset adjustment and external compensation options are available within the TL08x family. The C-suffix devices are characterized for operation from 0°C to 70°C. The I-suffix devices are characterized for operation from –40°C to 85°C. The Q-suffix devices are characterized for operation from –40°C to 125°C.

A.1.4 | Download Datasheet

<http://focus.ti.com/lit/ds/symlink/tl082.pdf>

APPENDIX D



System Lab Kit ASLKv2010 Starter Kit Connection Diagrams

Figure D-1 shows the overall floorplan of the ASLKv2010 Starter kit. We have shown the power connections in ASLKv2010 Starter in Figure D-2. Note that the $\pm 10\text{ V}$ power and ground connections have to be connected to the power inlets at the side of the kit; the power and ground is internally connected to the OPAMPs and analog multipliers. If you wish to carry out an experiment using the DAC integrated circuits on the board, you must use an external 5 V supply and ground connection, as shown in Figure D-2. There are three potentiometers included in the kit, each of which is connected across 10 V and ground. The output of the potentiometer can be used to derive a voltage in the range 0 to 10 V ; this can be useful in generating a reference voltage or even in generating a 5 V power supply for the DAC.

The student must become familiar with the general floorplan of the kit. Begin by understanding the power and ground connections to the kit. Note that $\pm 10\text{ V}$ and ground connections must be fed to the kit from the inlets at the left-hand side. This automatically powers the operational amplifiers and the analog multipliers. Each of the

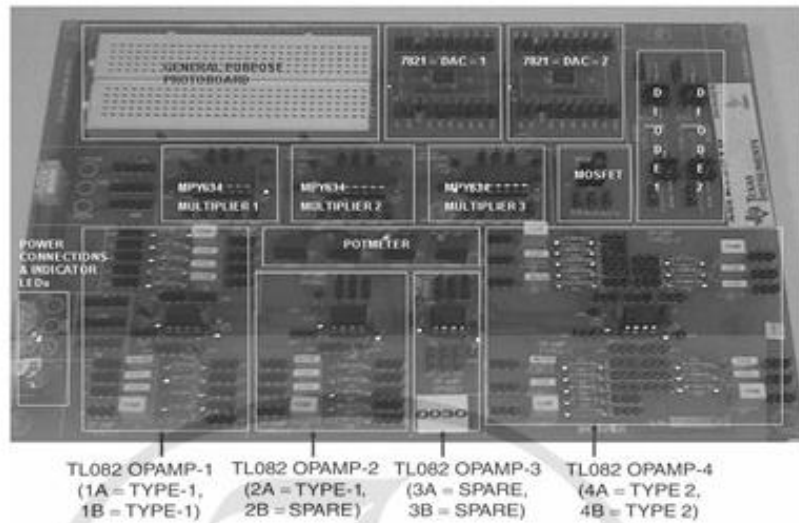


Figure D-1 Floorplan of the ASLKv2010 Starter kit

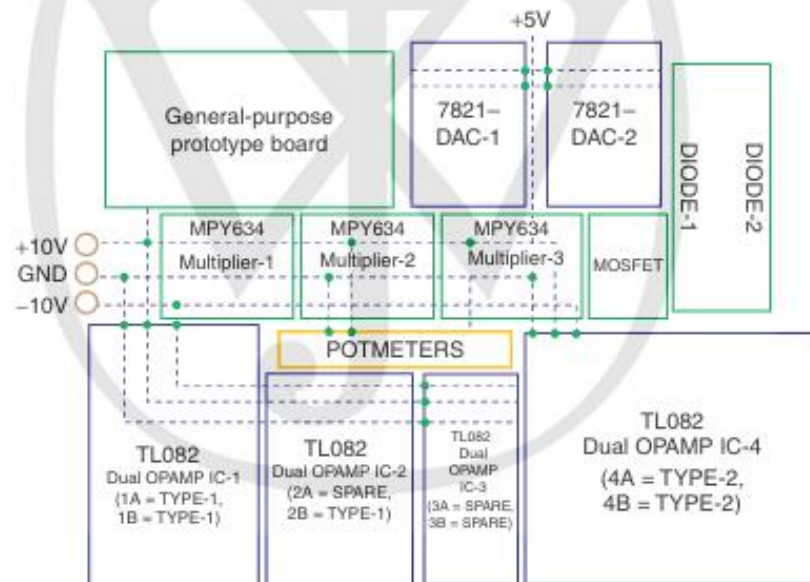


Figure D-2 Power connections in ASLKv2010 Starter kit. Only connect $\pm 10\text{ V}$ and ground connections

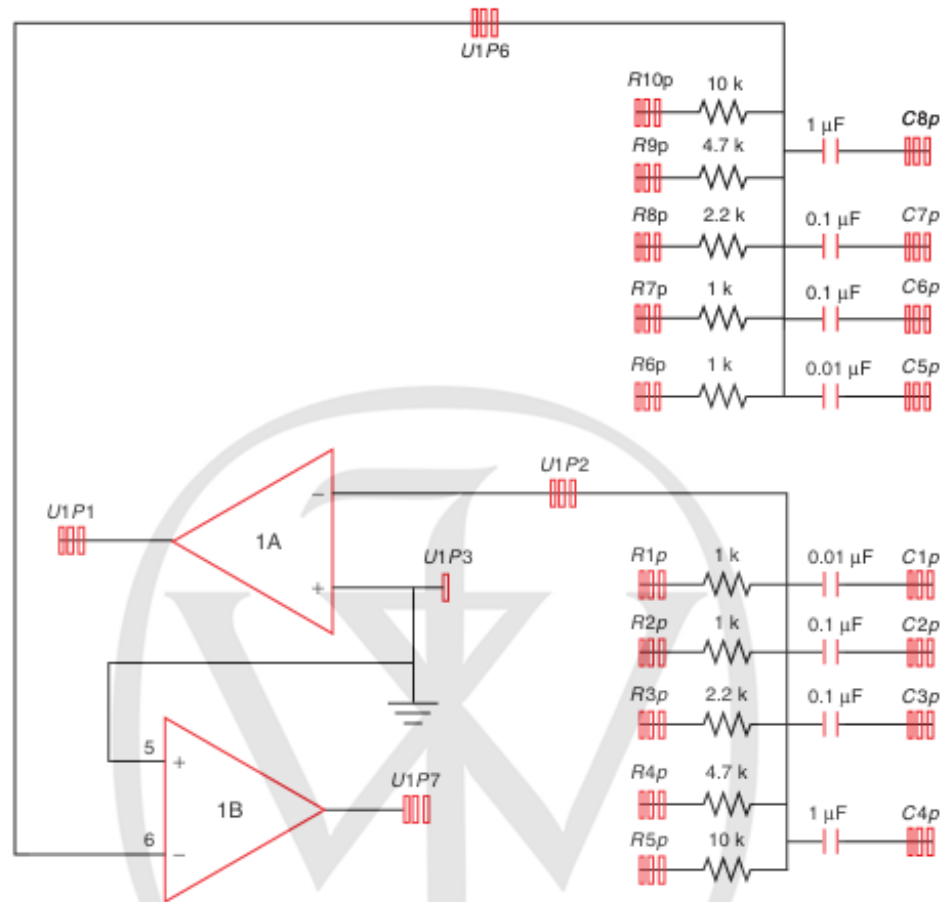


Figure D-3 Op-Amp IC-1 (Dual Op-Amp with two amplifiers, 1A and 1B) connected in Type-1 configuration (Inverting)

potmeters receives +10 V supply and ground connection. The output of a potmeter can be used as a DC reference voltage or even as power supply for the DAC.

Figure D-3 shows the connections for Op-Amp IC-1, which has two Op-Amps connected in Type-1 (inverting) configuration. The inverting terminal of Op-Amp 1-A is connected to resistors through Berg pin connections $R1p, R2p, \dots, R5p$ and

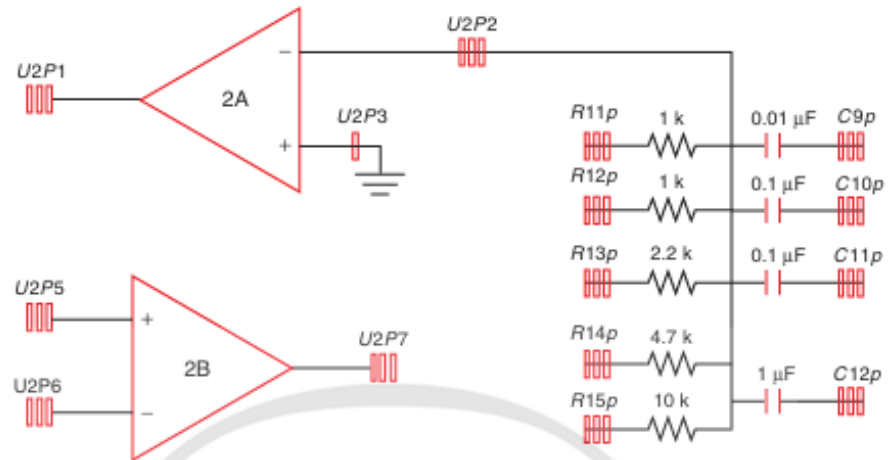


Figure D-4 Op-Amp IC-2 (Dual Op-Amp with two amplifiers, 2A and 2B. 2A can be connected in Type-1 configuration (Inverting; 2B is a spare)

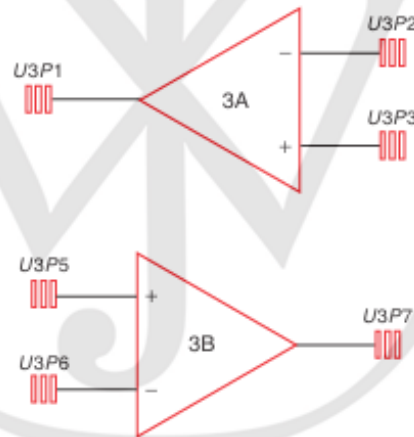


Figure D-5 Op-Amp IC-3 (Dual Op-Amp with two spare amplifiers, 3A and 3B)

to capacitors through the Berg pin connections $C1p, C2p, C3p, C4p$. Note that each Berg connection has three pins and the user can use any one of them for making an electrical connection. The Op-Amp 1-B is connected to resistors through Berg pin connections $R6p, R7p, \dots, R10p$ and to capacitors through Berg pin connections

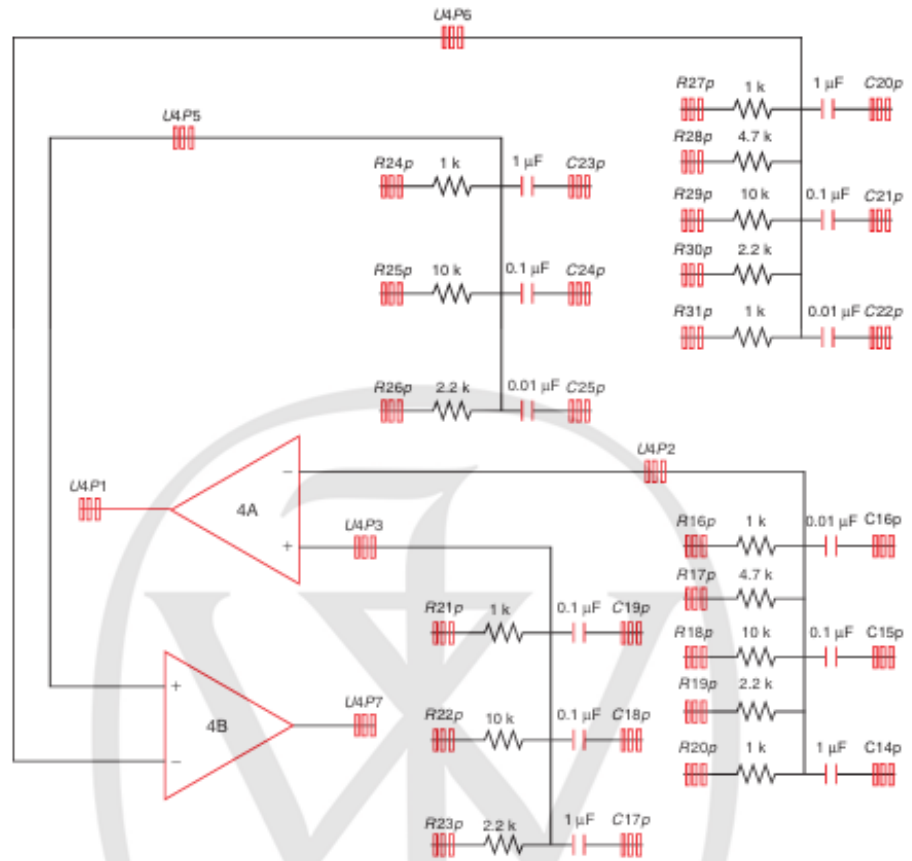
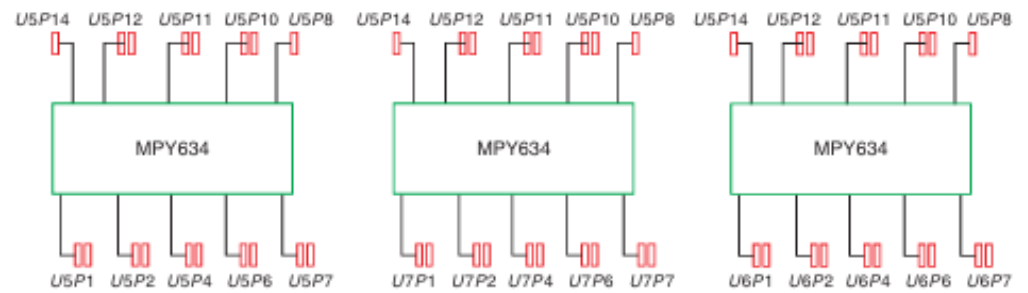


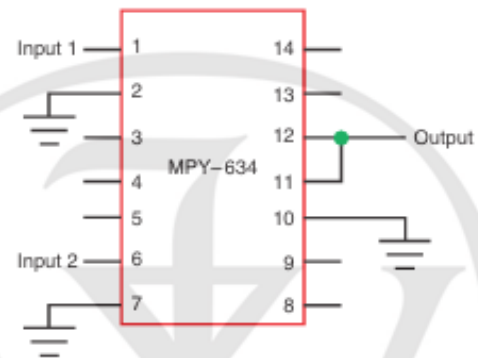
Figure D-6 Op-Amp IC-4 (Dual Op-Amp with amplifiers 4A and 4B. Op-Amp 4A and 4B can be used in inverting or non-inverting configuration)

C5p, C6p, C7p, C8p. The Berg pin connection *U1P1* can be used to connect the output of Op-Amp 1-A. Similarly, connection *U1P7* can be used to connect the output of Op-Amp 1-B.

Appendix D



(a)



(b)

Figure D-7 (a) PCB connections for analog multipliers 1, 2 and 3 on ASLKv2010 Starter; (b) External connections needed to use the multiplier MPY634

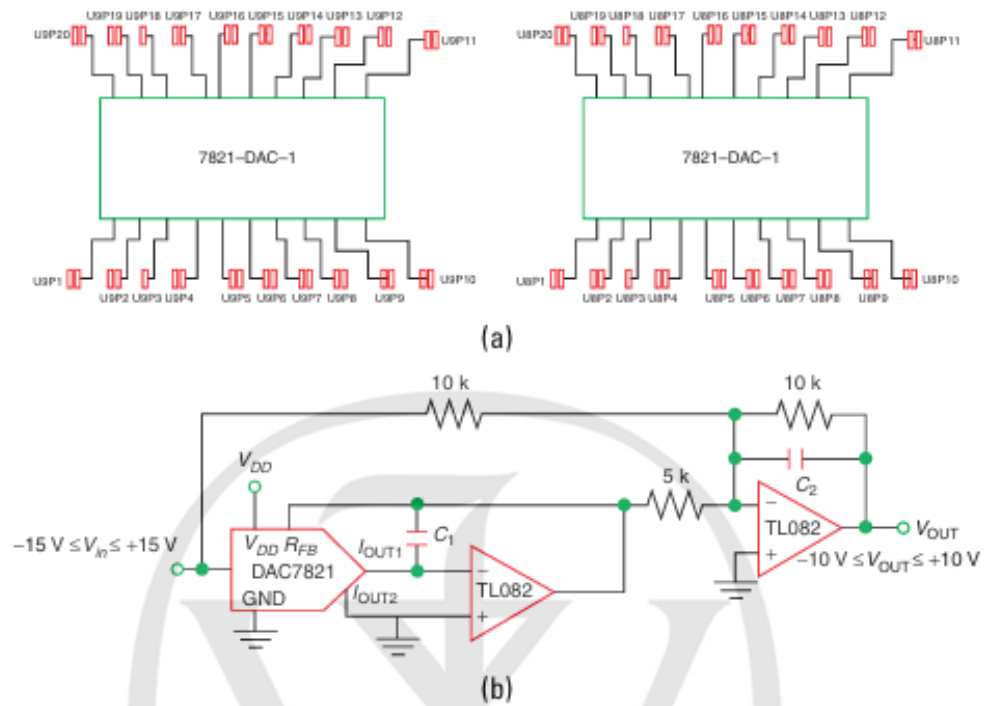


Figure D-8 (a) PCB connections for the DAC on ASLKv2010 Starter; (b) external connections needed to use the DAC to make it four-quadrant